

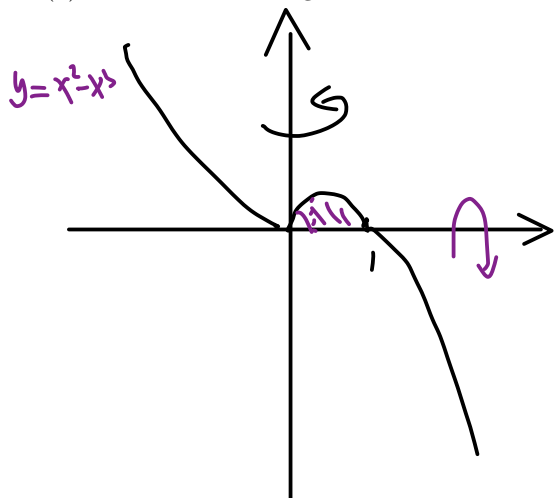
$$V = \int_a^b A(x) dx \quad \left\{ \begin{array}{l} \text{disk} \\ \text{washer} \\ \text{cylindrical shell} \end{array} \right. \quad \int 2\pi x f(x) dx ,$$

MATH 126

B2 past final worksheet

1. (Fall 2024, #4) Consider the region bounded by $y = x^2 - x^3$, and $y = 0$, where $x \geq 0$. Find an integral formula for the volume of the solid obtained

(a) if we rotate the region about the x -axis (do not calculate the actual value),



$$y = x^2 - x^3 = x^2(1-x)$$

$x \geq 0$.

Disk method

$$V = \int_0^1 \pi (f(x))^2 dx$$

$$= \int_0^1 \pi \cdot (x^2 - x^3)^2 dx$$

(b) and if we rotate the region about the y -axis (do not calculate the actual value).

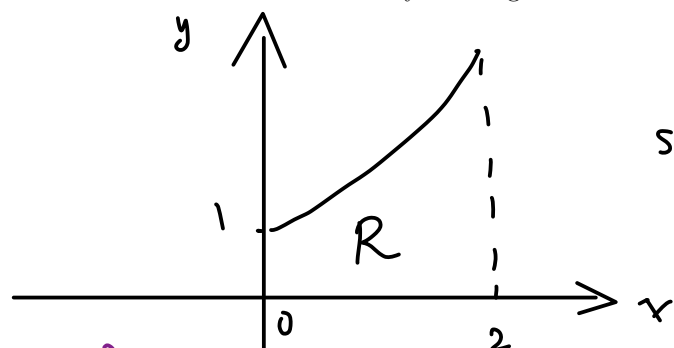
Shell Method.

$$V = \int_0^1 2\pi x f(x) dx$$

$$= \int_0^1 2\pi x \cdot (x^2 - x^3) dx$$

2. (Fall 2023, #4)

- (a) Let $\mathcal{R}$ be the region bounded by the x -axis and the curve $y = e^x + x$, where $0 \leq x \leq 2$. Compute the volume of the solid obtained by rotating $\mathcal{R}$ about the y -axis.



$$f(0) = e^0 + 0 = 1$$

shell method

Integration by parts

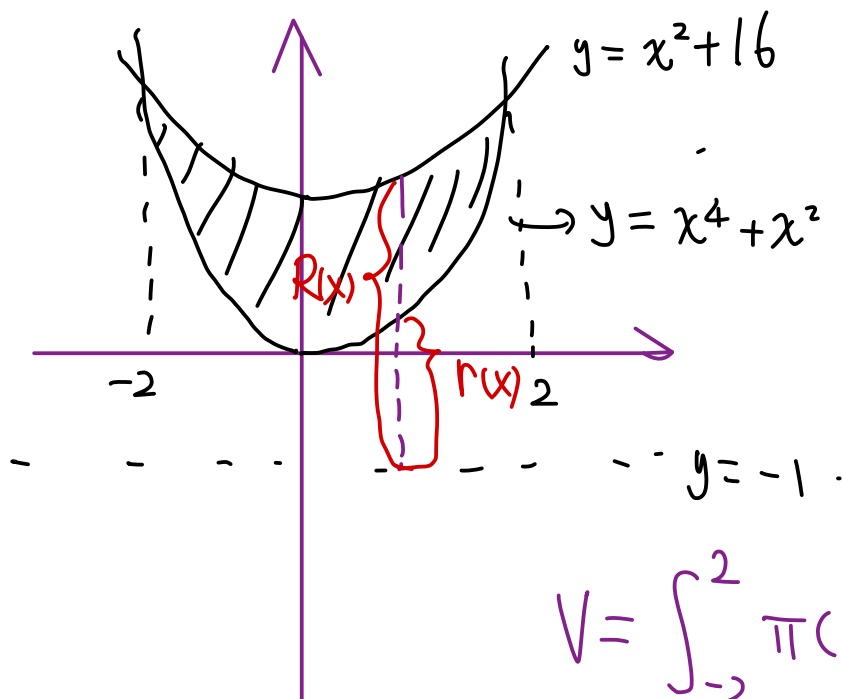
$$\begin{aligned} &= \int_0^2 2\pi x e^x + 2\pi \cdot \frac{x^3}{3} \Big|_0^2 \\ &= 2\pi x e^x \Big|_0^2 - \int_0^2 2\pi e^x dx \\ &\quad + 2\pi \cdot \frac{8}{3} \end{aligned}$$

$$\begin{aligned} &= 4\pi e^2 - (2\pi e^x \Big|_0^2) + \frac{16}{3}\pi \\ &= 4\pi e^2 - (2\pi e^2 - 2\pi) + \frac{16}{3}\pi \end{aligned}$$

$$= 2\pi e^2 + \frac{22}{3}\pi$$

$$\begin{aligned} V &= \int_0^2 2\pi x \cdot (e^x + x) dx \\ &= \int_0^2 2\pi x e^x dx + \int_0^2 2\pi x^2 dx \end{aligned}$$

- (b) The region $\mathcal{R}$ is bounded by the curves $x = 0$, $y = x^4 + x^2$, and $y = x^2 + 16$. Set up an integral expressing the volume of the solid obtained by rotating $\mathcal{R}$ about $y = -1$. You do not need to evaluate the integral.



$$\begin{cases} y = x^2 + 16 \\ y = x^4 + x^2 \end{cases}$$

$$\Rightarrow x^4 + x^2 = x^2 + 16$$

$$\Rightarrow x^4 = 16 \Rightarrow x = \pm 2$$

Washer Method

$$V = \int_{-2}^2 \pi (R^2(x) - r^2(x)) dx$$

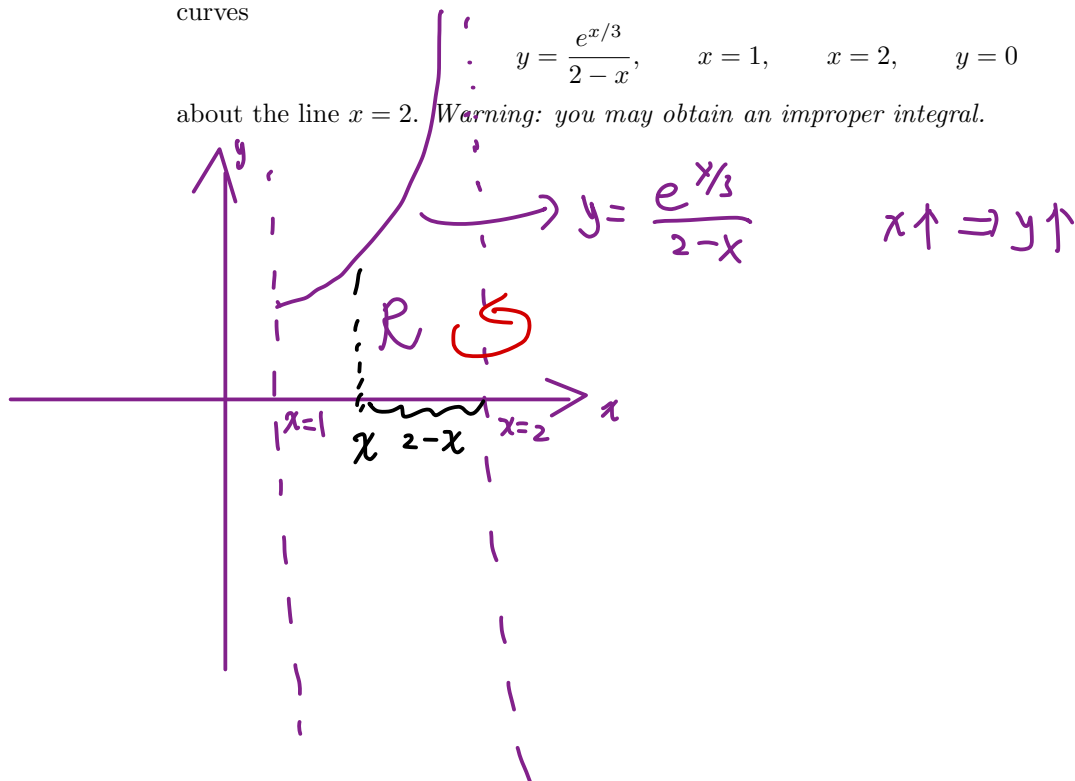
$$x=0$$

$$= \int_{-2}^2 \pi \left[(x^2 + 16 + 1)^2 - (x^4 + x^2 + 1)^2 \right] dx$$

3. (Spring 2023, #2) Find the volume of the solid obtained by rotation of the region bounded by the curves

$$y = \frac{e^{x/3}}{2-x}, \quad x=1, \quad x=2, \quad y=0$$

about the line $x=2$. *Warning: you may obtain an improper integral.*



shell method

$$V = \int_1^2 2\pi (2-x) f(x) dx.$$

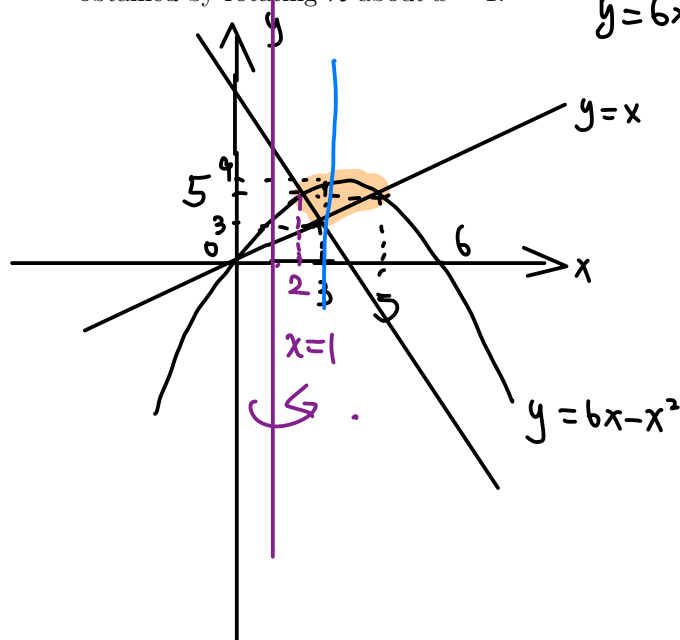
$$= \int_1^2 2\pi (2-x) \frac{e^{x/3}}{2-x} dx$$

$$= \int_1^2 2\pi e^{x/3} dx.$$

$$= 2\pi \cdot 3e^{x/3} \Big|_1^2$$

$$= 6\pi (e^{2/3} - e^{1/3}).$$

4. (Fall 2022, #5) Let $\mathcal{R}$ be the region containing the point $(3, 6)$ enclosed by the curves $y = 6x - x^2$, $y = x$, and $y = 18 - 5x$. Set up, but **do not evaluate**, an integral expressing the **volume** of the solid obtained by rotating $\mathcal{R}$ about $x = 1$.



Shell Method

$$\begin{cases} y = 6x - x^2 \\ y = x \end{cases} \Rightarrow x = 6x - x^2 \\ \Rightarrow x^2 - 5x = 0 \\ \Rightarrow x = 0 \text{ or } 5$$

$$\begin{cases} y = 18 - 5x \\ y = x \end{cases} \Rightarrow 18 - 5x = x \Rightarrow \begin{cases} x = 3 \\ y = 3 \end{cases}$$

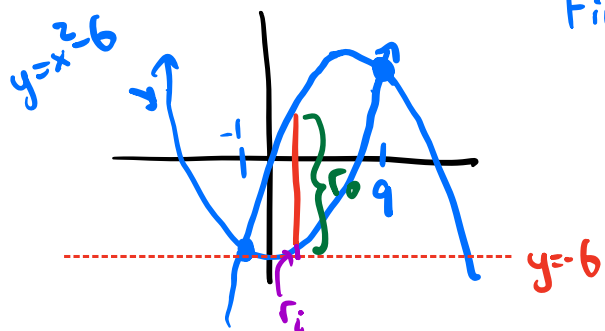
$$\begin{cases} y = 18 - 5x \\ y = 6x - x^2 \end{cases} \Rightarrow 18 - 5x = 6x - x^2 \\ \Rightarrow x^2 - 11x + 18 = 0 \\ \Rightarrow (x - 2)(x - 9) = 0 \\ \Rightarrow x = 2 \text{ or } x = 9$$

$$V = \int_2^3 2\pi(x-1)[(6x-x^2)-(18-5x)]dx \\ + \int_3^5 2\pi(x-1)[(6x-x^2)-x]dx$$

$$-(x^2 - 16x + 12)$$

5. (Fall 2021, #5) Consider the region R bounded by $y = x^2 - 6$ and $y = -x^2 + 16x + 12$. Set up, but DO NOT EVALUATE, an integral for the volume of the solid obtained by rotating the region R :

(a) about the line $y = -6$,

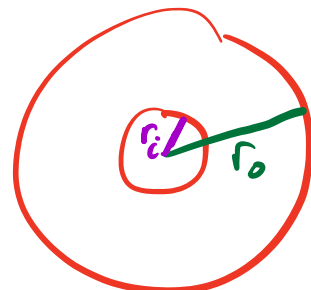


Find intersections:

$$x^2 - 6 = -x^2 + 16x + 12$$

$$2x^2 - 16x - 18 = 0$$

$$2(x+1)(x-9) = 0$$



$$r_o = -x^2 + 16x + 12 - (-6)$$

$$r_i = x^2 - 6 - (-6)$$

$$\int_{-1}^9 (\pi(-x^2 + 16x + 24)^2 - \pi(x^2)^2) dx$$

(b) and about the line $x = -6$.

Shells:

$$\int_{-1}^9 2\pi(x - (-6))(-x^2 + 16x + 24 - (x^2 - 6)) dx$$